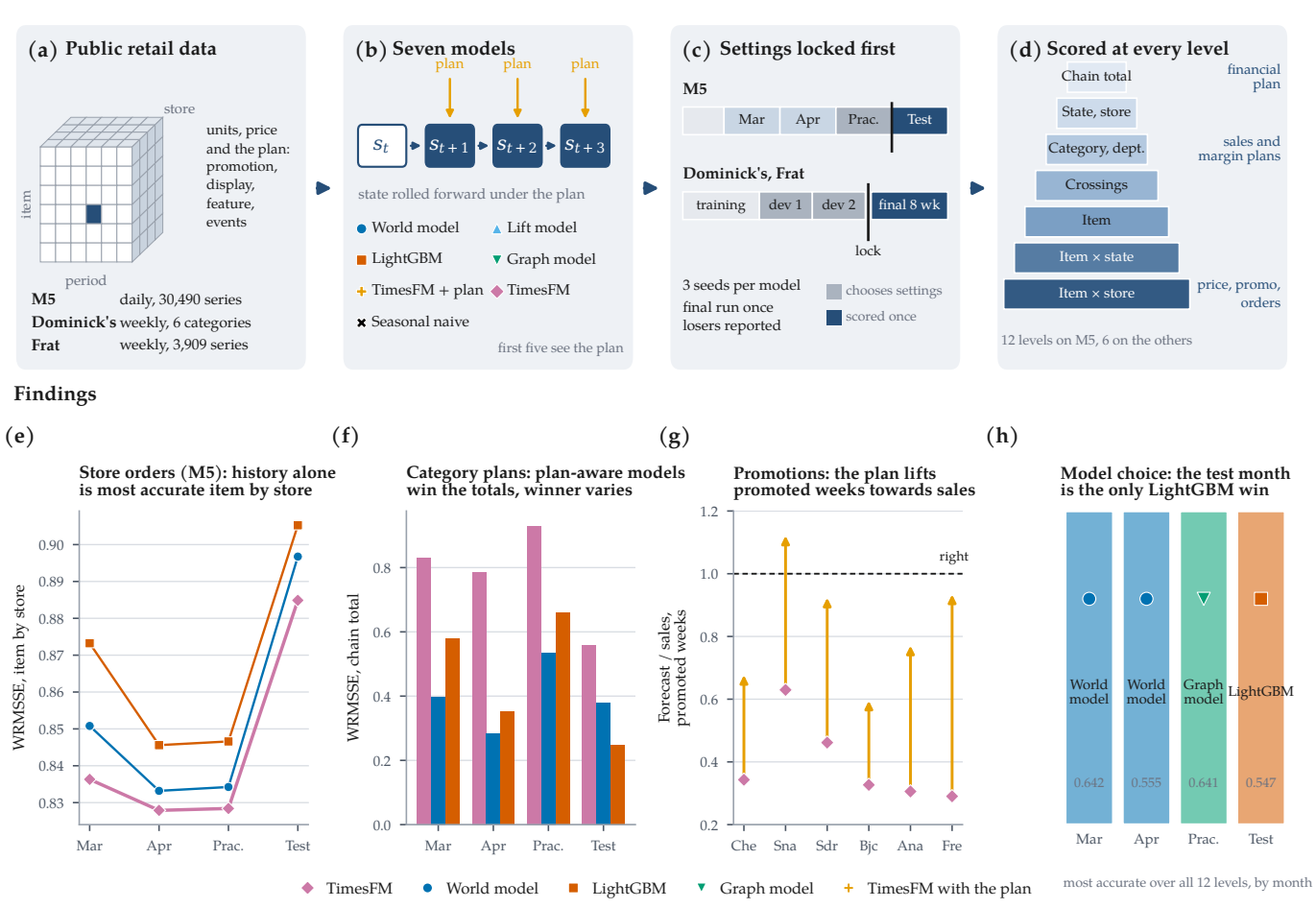
(a) Public retail data
store
item
period
units, price and the plan: promotion, display, feature, events
M5 daily, 30,490 series
Dominick's weekly, 6 categories
Frat weekly, 3,909 series
(b) Seven models
plan
plan
plan
$s_t$
$s_{t+1}$
$s_{t+2}$
$s_{t+3}$
state rolled forward under the plan
World model
Lift model
LightGBM
Graph model
TimesFM + plan
TimesFM
Seasonal naive
first five see the plan
(c) Settings locked first
M5
Mar
Apr
Prac.
Test
Dominick's, Frat
training
dev 1
dev 2
final 8 wk
lock
3 seeds per model
final run once
losers reported
chooses settings
scored once
(d) Scored at every level
Chain total
State, store
Category, dept.
Crossings
Item
Item × state
Item × store
financial plan
sales and margin plans
price, promo, orders
12 levels on M5, 6 on the others
Findings
(e)
Store orders (M5): history alone is most accurate item by store
WRMSSE, item by store
0.90
0.89
0.88
0.87
0.86
0.85
0.84
0.83
Mar
Apr
Prac.
Test
(f)
Category plans: plan-aware models win the totals, winner varies
WRMSSE, chain total
0.8
0.6
0.4
0.2
0.0
Mar
Apr
Prac.
Test
(g)
Promotions: the plan lifts promoted weeks towards sales
Forecast / sales, promoted weeks
1.2
1.0
0.8
0.6
0.4
0.2
right
Che
Sna
Sdr
Bjc
Ana
Fre
(h)
Model choice: the test month is the only LightGBM win
World model
World model
Graph model
LightGBM
0.642
0.555
0.641
0.547
Mar
Apr
Prac.
Test
most accurate over all 12 levels, by month
TimesFM
World model
LightGBM
Graph model
TimesFM with the plan